

Enhancing Sports Experiences Through Video-Based Interactions

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Abstract

Video is an essential part of sports interaction among sportspeople. Athletes benefit from self-visualization to correct movement patterns, coaches leverage video analysis to review past events and spectators engage with video to stay connected with their preferred sports. To meet these distinct user needs, it is crucial to establish a holistic approach that considers the intricacies of human interactions within sports contexts. This PhD research adopts a user-centered approach that iteratively creates, develops and evaluates intelligent video-based systems to support meaningful sports interaction. In line with previous work, this research emphasizes how these systems can facilitate visual analysis and knowledge sharing while addressing interaction challenges posed by deploying computer vision in real-world sports scenarios. The findings from this research contribute to the emerging field of SportsHCI, providing design implications for intelligent video-based systems that enhance learning, game analysis and overall interaction across sportspeople. Additionally, it recognizes the significance of machine learning methods in supporting interpersonal collaboration (e.g., athlete-coach), game understanding and personalization in video-based interactions. Situated at the intersection between HCI, computer vision and SportsHCI, this work aligns with core tenets of research in the field that contribute to enhancing user experiences across multimedia applications.

CCS Concepts

• Human-centered computing → Interaction paradigms; Empirical studies in HCI.

Keywords

Video; HCI; Multimedia; Sports; SportsHCI; Computer Vision; Machine Learning; Ubiquitous Systems

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1 Introduction

The consumption of multimedia across online platforms has become a significant and ubiquitous part of how people engage with their

preferred content in their regular everyday interactions. In this space, video presents itself as a versatile media that can reach global audiences while still being capable of supporting meaningful interactions at individual and community levels. Sports are a notable example of this dichotomy, as large sports organizations benefit from disseminating video content to their audience, while coaches can leverage video recordings to review key moments and boost information retention for athletes [12, 22].

Integrating technology into sports practices is a well-represented topic in the literature, playing a significant role in how sports training is becoming increasingly more analytical and intelligent. Research has explored a diverse set of technologies to gain additional knowledge directly from sports training such as wearable devices (e.g., smartwatches) and specialized sensors (e.g., inertia and pressure sensors) [15, 16]. Such approaches are essential to complement our understanding of the complex interactions and dynamics present in sports. However, research often neglects to study the nature of existing sports interactions and, thus, fails to fully understand the resources available to athletes, coaches and recreational sportspeople. For instance, a frequent situation in collective sports training is the limited (possibly shared) space and time available for athletes and coaches. Similarly, regulatory constraints in official regulations often prohibit the use of specific equipment during competition (e.g., wearables) with the intent of safeguarding the health of their athletes [18]. Therefore, video stands out as a non-intrusive, widely accessible resource that can be leveraged without interfering with the natural flow of training and competition. This makes it an ideal research tool, which perfectly aligns with this research area of intervention at the intersection of HCI, multimedia and computer vision.

Raw video recordings already provide sports stakeholders with the means to document, review and engage with previous events. These processes can promote collaboration between athletes and coaches, who can benefit from analyzing visual cues to understand and improve aspects relevant to their craft. Even though these advantages are well-established, existing research struggles to provide a comprehensive understanding of how to support the coach-athlete relationship using interactive video technology [8]. This is a core objective of this research.

To extend the inherent interactions with video data, computer vision techniques provide opportunities to expand analytical depth. Vision techniques such as pose estimation and object detection make it possible to extract meaningful insights from standard monocular video, which can support additional analysis of biomechanics, spatial interactions, and related contextual information [9]. However, leveraging vision models in sports environments presents a challenging task. Most existing approaches are primarily optimized for low-level object recognition and fall short in capturing



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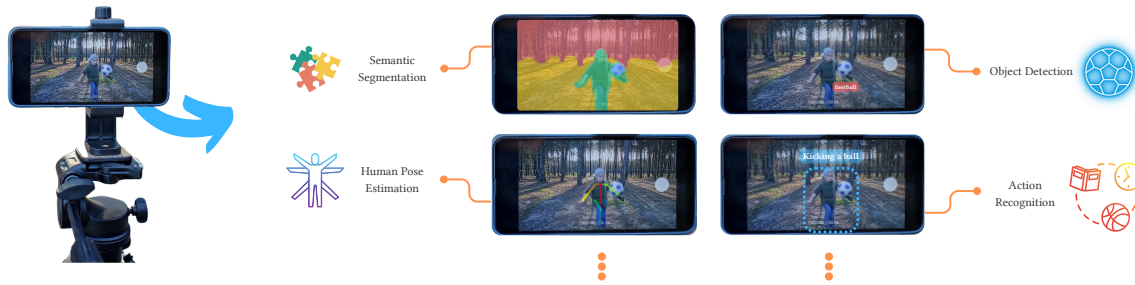


Figure 1: Examples of computer vision models supporting video-based features in the soccer context. Top left - semantic segmentation, bottom left - human pose estimation, top right - object detection and bottom right - action recognition.

high-level semantic or contextual information, such as tactical outcomes or interaction effects, which are crucial for sports analysis [20]. Additionally, models often struggle with low-quality or broadcast footage, where factors such as low frame rates and occlusion hinder accurate detection and tracking of fast-moving objects. Research in this field often aims to address specific vision challenges arising from the visual complexity of standard sports video recordings [1]. For instance, recent research efforts aim to suppress the inaccurate inferences resulting from the effects of motion blur - a frequent side-effect in most sports-related video - which makes it significantly harder to track elements in most sports [6]. These improvements are crucial as they enable computer vision to become an additional resource that can positively impact the ideation and development processes and create robust functionalities. Thus, the challenge lies in adapting to such limitations and preparing system interactions to provide positive user experiences.

There are noteworthy systems that already demonstrate how computer vision can be effectively applied to sports video applications. In this context, video navigation is a standard interaction that enables sportspeople to search for specific moments to review. Besides providing additional visual insights (e.g., pose overlays, movement correction), the action of finding them can present a time-consuming process for athletes, coaches and spectators [7]. However, research tends to address this need indirectly, primarily through video summarization techniques or highlight detection [21]. These approaches provide valuable overviews of game content but often overlook the development of interactive tools that enable users to precisely locate and navigate to specific moments of interest within sports videos. This represents an underexplored area of research - particularly within the field of SportsHCI - that stands to benefit from studying navigation methods that align with the needs of sports stakeholders. Such methods include features for manual annotations, event tagging or audio cue-based search to support more efficient and targeted video exploration [3, 5].

This research adds to the emerging interdisciplinary domain of SportsHCI following a user-centered approach that has iteratively designed, developed and tested video-based systems within sports. By combining computer vision outputs with video-based interactions, subsequent developments aim to positively impact our understanding of interactive technologies across sports, providing actionable insights and solutions that are both technologically robust and contextually sensitive to real-world scenarios. Therefore, to enhance sports experiences through video-based interactions,

this work makes the following contributions: 1) A comprehensive framework for video-based athlete-coach interaction systems, 2) Novel computer vision integration techniques for real-time sports video analysis, 3) Empirical validation through iterative user studies across multiple sports contexts, 4) Design implications for intelligent video systems in ubiquitous sports environments.

2 Related Work

The development of interactive technologies within the Sports Human-Computer Interaction (SportsHCI) research is essential to supporting physical activities among athletes, coaches and recreational sportspeople. To understand the motivation behind this work it is necessary to introduce the contributions that guide current research efforts. Hence, the upcoming subsections provide a brief description of the reference literature that supports the methodology of this work.

2.1 Interaction Research

Sports encompass a diverse set of disciplines that naturally result in vastly different contexts, each with inherent unique characteristics. Thus, an essential first step is to consider a suitable methodology that optimizes technological interactions with sports stakeholders effectively. Research Through Design [29] provides a foundation for HCI research to generate optimal solutions, with the purpose of balancing research and creation to develop a research prototype that can represent a research contribution. Another core motivation stems from research in the emerging field of SportsHCI, which explores interaction design to support physically active individuals. Notably, Elvitigala et al. [8] identify grand challenges in SportsHCI, specifically highlighting the need for better athlete-coach relationships as a core challenge area.

While near-perfect inference results are ideal, for the interaction design to be successful, less-than-perfect Machine Learning outputs might suffice. Research has contributed to providing interaction design patterns that enable users to interact effectively by addressing the trade-off between performance and user experience [23]. For instance, search engines (e.g., Google Search) present a list of top-ranked links instead of directing the user to a single page. As Yildirim et. al. [25] describe, many failures result from neglecting human-centered design early in the innovation process. Such design considerations are crucial as they enable and shape human-centered interaction.

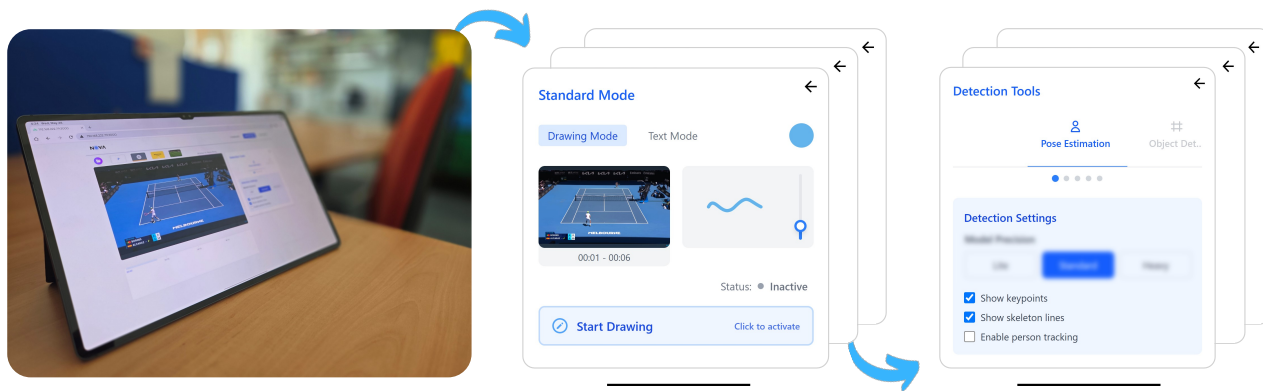


Figure 2: Tablet view of the web-based prototype displaying the main interaction page (left interface) with additional side panels for video interaction (middle and right).

2.2 Sports Systems

Multimedia is an essential resource for promoting positive interactions that help increase our communication capacity and drive audience interest. In sports contexts, analyzing video recordings supports learning across multiple activities, impacting individual sports - where performance rests heavily on a single athlete - as well as team sports, which also benefit on a large scale from spatial-temporal analysis of players during the games to find or perfect new strategies and tactics [4, 10, 11].

Recent advances in computer vision have generated various options for achieving reliable video processing (Figure 1). These improvements have led to the adoption of modern architectures (e.g., transformer-based) that outperform traditional ones [19, 26]. Naturally, this has also impacted how specialized motion-sensing devices, such as the Kinect, has been increasingly replaced by solutions that integrate deep-learning architectures requiring only standard images/video frames as input [13].

Several sports-driven systems have inspired this research. On one hand, there are video-based annotation systems that have employed a diverse set of digital note types (e.g., drawing, text and images) in contexts such as contemporary dance [17]. In parallel, the need for intelligent features capable of supporting video analysis constituted a central goal for this PhD work. Video analysis has become crucial for evaluating player performance and strategizing gameplay by applying advanced detection and tracking models to movement data. Systems such as EventAnchor [7] leverage the timing of events to analyze gameplay videos to automatically detect key moments. Another example is VisCommentator [28], which enables sports analysts to enhance visualization with augmented videos, making it easier to communicate complex gameplay insights.

There are numerous examples of systems targeting sports interaction, often framed by a case-study nature that allowed research to tackle specific challenges in each sport [24, 27]. When developing research solutions that improve interaction aspects in sports, understanding that such solutions must operate within real-life constrained systems to tackle an existing need is imperative [14]. Hence, the contributions made in the work *Grand Challenges in SportsHCI* [8] are especially noteworthy, particularly by addressing

the role of interactive technologies in supporting context-aware, responsive interactions across diverse sporting roles and environments.

3 Methodology and Work In Progress

The methodology employed in this work follows a Research Through Design approach that makes prototyping a tool for inquiring into design knowledge. This methodology enabled our research efforts to extract further insights into both previous and current system developments. Insights were guided by user-centered feedback on the applicability of system features within their ubiquitous environments (e.g., basketball training and dance rehearsals). This iterative approach builds upon a series of prototype developments that collectively inform the contributions presented in this thesis.

3.1 Previous Work

During the initial development of our web-based video annotation system, a central research goal was to evaluate the applicability of multimedia annotations across ubiquitous environments. This system allowed for general annotation capabilities such as drawings and text overlaying a video player. Existing functionalities were designed to facilitate video archiving, documenting and reviewing information. Following an initial iteration, the system received valuable feedback from participants and the HCI community, motivating a novel system iteration that adapted existing human movement analysis to a visually complex sports setting. A key takeaway during the evaluation of this system was the role of video-based features in stimulating self-visualization and supporting subsequent movement correction. Thus, an early research objective of this PhD was to study how video-based interaction could positively support the athlete-coach relationship. Interestingly, our initial goal ultimately aligned closely with the second grand challenge proposed in the work of Elvitigala et al. [8].

These initial experiments were crucial to understanding the overall positive impact of video-based systems on the physical activities of sportspeople. With this in mind, the video annotation system integrated pose estimation, allowing for the additional visualization of body movement to overlay a video. The resulting system

iteration was tested on basketball and contemporary dance case studies, with participants sharing encouraging remarks and rating their interactions positively. Functionalities such as dynamic annotations — annotations capable of following selected keypoints (body joints) — were a noteworthy feature during this phase.

3.2 Ongoing Developments

In the work surrounding this annotation system, functionalities were designed for general-purpose interactions, enabling users to create their intended annotations. In contrast, features derived from vision models primarily target video analysis for individual body movements. Sports interactions benefit from insights beyond technical analysis, such as tactical and strategic decision-making, which are essential for comprehensive sports analysis. Thus, to initiate our research into more general scope interactions with sports, our latest work focused on developing a proof-of-concept system that leveraged object detection and tracking to create functionalities that would support practitioners of racket sports.

Sports-driven systems, such as EventAnchor [7] and AiCommentator [2], demonstrate how computer vision can be effectively applied to sports video applications. Their work motivated another significant research objective of this work, focusing on a key aspect of standard video interaction that frequently occurs during coach-athlete video interactions: timeline navigation. In previous work, our prototype enabled users to navigate a selected video by jumping to a timestamp containing a previously created annotation (e.g., a drawing or text). However, depending on the intended action, there are numerous potential ways to search through a video to find a specific timestamp, such as a game event (e.g., a goal), an action (e.g., a ball throw) or commentary. These interactive options constitute another research focus aimed at evaluating how key stakeholders in sports — such as coaches and analysts — perceive, interact with and benefit from timeline-based functionalities in the context of video analysis.

To unify previous research workflows, a new prototype is in development. The feedback received throughout testing sessions and peer reviews has informed the design choices for this system's interaction and technological backbone (Figure 3). In parallel, to continue to answer the proposed research questions, our research group is organizing interview phases with PE instructors for upcoming user studies. This participant category represents a valuable research asset given their regular contact with diverse sports disciplines. Moreover, they often navigate resource-constrained environments inherent to real-world settings, pertinent to this discussion topic. This serves to guide developments into extending our computer vision capabilities for analysis across sports contexts effectively.

4 Ethical Considerations and Reproducibility

Even though we do not anticipate major ethical concerns in the proposed research plan, our research group identifies the following possible ethical outliers: physical exertion during testing sessions, personal user data (e.g., videos) used for academic purposes and potential research with underage participants such as young athletes. To address foreseeable and unforeseen ethical considerations, the research will strictly adhere to the guidelines and authorization procedures established by the Ethics Committee of the NOVA

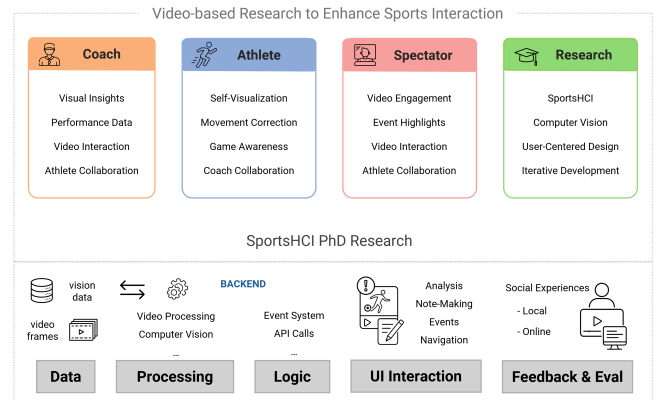


Figure 3: Schematic of the main topics of this research work as well as methodological and development pipelines.

School of Science and Technology (CENOVAFCT). The committee will provide oversight and guidance throughout the PhD process to ensure that data collection, participant recruitment and testing protocols adhere to informed consent, data protection regulations, and broader ethical standards.

To ensure the reproducibility and transparency of this work, the developed web-based system, including other auxiliary tools and practical components, will be made freely available to the research community. Additionally, all relevant source code and technical implementations will be shared via an open-access GitHub repository. Empirical results, including user feedback, usability metrics and qualitative insights collected over the course of this research, will be systematically documented and published following the methodological standards established in this work.

5 Conclusion and Future Work

This research studies advanced vision algorithms and video processing to contribute to the field of Human-Computer Interaction (HCI). By complementing video modalities with automatically extracted context-dependent data, this work will contribute to assisting athletes, coaches, spectators and recreational sportspeople with intelligent sports interaction. The methodology for this PhD research follows a user-centered approach that can guide the development of the existing video-based system.

The findings from this work result from integrating computer vision techniques iteratively to enhance video-based sports interactions while effectively addressing real-world challenges. Through iterative prototyping and user feedback, this research contributes actionable design implications for intelligent video systems that support learning and engagement, impacting athlete-coach interactions and sports organizations.

Future work will focus on expanding the system's capabilities to support advanced navigation methods that align with the specific contextual needs of stakeholders. This research advances the emerging SportsHCI domain by providing interactive vision systems and design patterns that bridge computer vision capabilities with human-centered sports experiences.

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